

The Shape of the Theory

A narrative entry to a programme that derived physics from nothing.

Mark E. Mala (pen name of Ekram Alam) **June 2026**

Abstract

This is the doorway paper to a research programme of 22 published papers, three custom AIs, and a Lean 4 / Mathlib infrastructure of ~90 files. The programme proposes a Theory of Everything that begins with **Nothing** and ends — *as a structural proposal, partially mechanised* — with the Standard Model, General Relativity, and quantum mechanics. A core of foundational results is genuinely machine-verified against Mathlib (the Lawvere fixed-point construction, seed selection, Lie-algebra rank-nullity, Clifford-matrix isomorphism, finite-dimensional Hilbert structure). A larger body of downstream physical claims is currently scaffolded in Lean but not yet end-to-end proved; closing that gap is ongoing work, and every claim in this paper wears a status tag that says exactly where it sits. This document is the narrative version: ten pages, no formalism unless it earns its keep, written for an intelligent generalist rather than a specialist. The formal papers (D, E, F, and the full Tree of Reality) carry the mathematics. This one carries the shape.

Everything below is open, falsifiable, and verifiable. Every paper has a permanent Zenodo DOI. Every commit is Bitcoin-timestamped. Every claim wears a status tag — what is proved, what is partial, what is predicted, what is open — because the point of doing this kind of work in the open is to be checked.

1. The Question

What is reality made of?

Not metaphorically. Not philosophically. **Mechanically.** What is the generative process? If you started with nothing — and then nothing forced itself into being — what would the rest of physics have to look like?

For most of the last hundred years, the standard answer has been: *we don't know, but we have very good descriptions of the parts*. General relativity describes gravity beautifully. Quantum mechanics describes the small beautifully. Neither agrees with the other at the seams. The Standard Model contains roughly nineteen free parameters nobody can derive from first

principles. The cosmological constant is wrong by a hundred and twenty orders of magnitude. The questions are real, and they have been sitting in the open for a long time.

This programme takes the opposite tack: assume there is a generative mechanism, and try to find it. Start from Nothing. Let self-reference do the only thing it can do. See what falls out.

What falls out, surprisingly, is most of physics.

2. How This Was Made

Before any of the substance: a note on who built it, and how.

This is the work of one person — me, writing under the pen name Mark E. Mala — in collaboration with frontier AI. No lab, no institution, no funding, no team. Built on the side for the love of the question.

The collaboration is structural, not cosmetic. Three custom AI systems were built for it:

- **Gnosis** — a discovery system that surveys results across distinct scientific fields and looks for deep structural patterns: not surface similarities, structural matches at the level of generative form.
- **Logos** — a formalisation system that takes informal mathematical claims and attempts to render them as machine-verified Lean 4 proofs against Mathlib.
- **Synthesis** — originally a composition system; subsequently retired. Papers are now composed manually using Claude Code, because the higher-rigour parts of the work benefit from a human in the writing loop.

The four-component architecture is described separately in *Pansophia* (Zenodo: [10.5281/zenodo.19974680](https://zenodo.org/record/105281/versions/19974680)). The fourth component, **Praxis**, is application — the eventual use of what's discovered. That comes later.

The substance below — the construction, the equations, the predictions — is the output of running these systems together over a long period, with constant human direction and verification. Some of what came back was extraordinary. Some of it was wrong, and got thrown away. Some of it is in this paper with honest tags so you can see exactly which is which.

One specific honesty about the Lean infrastructure deserves stating up front, before the status tags in §5 do their work. The Lean codebase exists, builds clean, and contains thousands of compiled facts. But an adversarial internal audit in early 2026 found that a large fraction of the downstream physics theorems were arithmetic proxies or type-level tautologies — *technically* compiling, but not actually establishing the physical content their docstrings claimed. A targeted upgrade pass moved some of these to genuine Mathlib proofs (the Lie-algebra rank-nullity work, the Clifford-matrix isomorphism, the cascade's dimensional structure). Others remain

scaffolded, and the project was paused in May 2026 when the genuinely hardest theorems (Bakry-Émery on non-compact manifolds, GNS reconstruction, L^2 spectral theory) were judged beyond current AI capability without human mathematician input. The status tags below reflect that honestly. The work continues.

I say this not as a hedge but as orientation. A theory written by one person and an AI collaborator deserves more scrutiny than one written by twenty physicists at a major lab. The provenance trail (Section 8) is designed to make that scrutiny easy.

3. The Seed: $D = (D \rightarrow D)$

The starting move is the smallest possible one.

Begin with **Nothing**. Ask: what is the absolute minimum a “thing” could be that would force itself to exist?

The answer, due originally to Dana Scott and crystallised by F. William Lawvere, is a **reflexive domain**: a structure that is identical to the space of maps from itself to itself. Symbolically:

$$D = (D \rightarrow D)$$

Read aloud: *D is the same as the collection of all ways to go from D to D*. The structure is its own function space. It refers to itself completely.

This is not a trick. It is the unique closed self-referential loop. If you try to make a structure that does not contain its own function space, you get classical sets, and you run into the Cantor / Tarski / Russell paradoxes. If you make a structure that *does* contain its own function space, in the reflexive-domain sense, the paradoxes dissolve — there is no “outside” position from which to formulate them. The loop is closed.

Why this matters for “from nothing”: the loop is atemporal. There is no “before” the reflexive domain. The closed self-referential structure *is* what makes existence possible at all. To ask “what triggered the cascade?” is to ask the wrong question — like asking what is north of the North Pole. The cascade is what self-reference looks like when you let it run.

From this loop, a minimum implementation has to be selected. We need an algebra that is

1. **Minimal** — the smallest non-trivial choice
2. **Non-commutative** — without it, you only get classical geometry, and quantum mechanics never appears
3. **Finite-dimensional** — so the cascade can be inspected directly
4. **Trace-faithful** — so probability and inner products are well-defined

Exactly one algebra satisfies all four constraints: $\mathbf{M}_2(\mathbb{C})$, the 2×2 matrices over the complex numbers. This is the seed.

Drop any one constraint and you get a different seed — quaternionic, octonionic, infinite-dimensional. The *space* of alternative seeds is real and not yet fully mapped (it is marked [SPECULATIVE] in the Tree of Reality and remains an open frontier). For our universe, the four constraints together pick $\mathbf{M}_2(\mathbb{C})$ uniquely under current understanding.

4. The Cascade and Three Lineages

From the seed, the construction has only three operations available — and each one generates a separate lineage of physics. This is the key claim of the theory, and it is the part most people find startling.

The cascade itself. Take `End` — the endomorphism operation, “all the linear maps from this thing to itself.” Apply it to $\mathbf{M}_2(\mathbb{C})$:

- `End( $\mathbf{M}_2$ ) =  $\mathbf{M}_4(\mathbb{C})$` — the 4×4 matrices
- `End( $\mathbf{M}_4$ ) =  $\mathbf{M}_{16}(\mathbb{C})$` — the 16×16 matrices
- And so on, indefinitely, but with one decisive moment at $\mathbf{M}_{256}$

The dimensional structure of this cascade is a genuine Mathlib result — `Module.finrank_matrix` plus `Fintype.card_fin` give `$\dim(\mathbf{M}_n(\mathbb{C})) = n^2$` , and the endomorphism step is then a straightforward algebraic fact. [PROVED for the dimensional cascade; PARTIAL for the full semantic claim that the cascade is the unique forced continuation from the seed.]

The branching. From the seed $\mathbf{M}_2(\mathbb{C})$, three distinct operations can be taken — and there are only three available, by the structure of the algebra itself:

Operation	What it gives	Lineage
<code>End</code>	observable algebra, matter content	END lineage
<code>Aut</code>	symmetry group, gauge structure	AUT lineage
<code>$\langle \cdot, \cdot \rangle$</code>	inner product, dynamics	$\langle \cdot, \cdot \rangle$ lineage

Each lineage develops independently. They do not merge.

The metaphor that fits is **evolutionary**, not architectural. Eyes evolved independently at least eight separate times in the history of life on Earth — cephalopods, vertebrates, jellyfish, certain worms, certain spiders. Each branch evolved its own camera eye on its own lineage. The branches never merged. The fact that all the eyes look broadly similar is **co-evolved similarity**

— not convergence, not merger. Just the same structural problem being solved by separate processes that share a common origin.

The three lineages of physics here behave exactly the same way. Each independently develops the structures it needs. The fact that all three produce (a version of) a Connes spectral triple, for instance, is co-evolved similarity. There is no merger node where the lineages collapse together. The similarity is a recognisable *pattern*, not a connection.

This matters because the older v2 of the framework spoke about “convergences” between branches, which implied merger. v3 retired that language entirely. Branches that have split do not converge. They may, separately, develop similar traits.

5. What Comes Out

This is where the theory either earns its keep or dies. We have a seed and three lineages. Do the right things actually fall out?

The answer, as far as the programme has gone, is *most of them*.

The END lineage (matter, observables):

- **4-dimensional spacetime as the cascade’s first non-trivial Clifford layer.**

`M4( $\mathbb{C}$ )  $\cong$  Cl4( $\mathbb{C}$ )` is a genuine algebra isomorphism, mechanised in Lean (`F4_1e_CliffordMatrix`). The structural claim that this *forces* physical 4D spacetime rather than merely matching it is the next step beyond the isomorphism. [PROVED for the isomorphism; PARTIAL for the forcing.]

- **Standard Model gauge group embedding.** The dimensional content `dim(su(3)) + dim(su(2)) + dim(u(1)) = 8 + 3 + 1 = 12 < 15 = dim(sl(4))` is genuine — proved via rank-nullity on the trace map in `TracelessMatrix`. The full Connes-classification claim that the SM is the *unique minimal* such embedding rests on the Chamseddine-Connes axioms and is currently scaffolded rather than end-to-end proved. [PARTIAL.]

- **Three generations of matter.** `dim(Im  $\mathbb{H}$ ) = 3` is a Mathlib fact; the fermion-space dimension `3 × 32 = 96` follows from `Fintype.card_prod`. The identification of this 3 with the physical generation count is structural, not derived from independent physics. [PROVED for the algebraic count; CLAIMED for the physical identification.]

- **Pati-Salam unification** at higher energy, with right-handed neutrinos forced. [CLAIMED.]

- **The Weinberg angle** comes out at `sin2θW = 3/8` at the Pati-Salam scale, running to ≈ 0.231 at `MZ` under standard RG flow. [CLAIMED for the 3/8 value as a structural consequence of the Pati-Salam embedding; PREDICTED for the `MZ` number.]

The AUT lineage (symmetry, forces):

- $GL_2 \rightarrow SL_2(\mathbb{C})$, and $SL_2(\mathbb{C}) \cong Spin(3,1)$ — the Lorentz group emerges as the AUT of the seed. [PARTIAL.]
- $Diff(M) \times Gauge(M)$ — diffeomorphism invariance plus gauge symmetry comes from the automorphism algebra of the full triple. [PREDICTED.]
- This terminates in **Einstein’s field equations** via Lovelock’s theorem (the unique second-order field equation in 4D is Einstein–Hilbert). [PREDICTED.]

The $\langle \cdot, \cdot \rangle$ lineage (geometry, dynamics, QM):

- A finite-dimensional Hilbert space structure on $\mathbb{C}^2$ with a canonical inner product. [PROVED.]
- **The Born rule** follows from Gleason’s theorem applied to the spectral measure. Gleason’s theorem itself is established literature; its application to the cascade’s spectral measure is cited rather than mechanised here. [CLAIMED via cited theorem.]
- **The Schrödinger equation** follows from Stone’s theorem applied to the unitary evolution. [PREDICTED.]
- **Pauli exclusion** from the antisymmetric tensor on the finite Hilbert space. [PREDICTED.]
- **No-cloning, no-deleting, no-broadcasting** — all derived in the literature from linearity and unitarity; not separately mechanised in this codebase. [CLAIMED.]

The spectral action then ties it back together:

$$S = \text{Tr}(f(D^2/\Lambda^2)) + \langle \psi, D\psi \rangle$$

This single object, expanded as a heat-kernel series, produces **gravity** (at order Λ^2), **Yang-Mills field strength F^2** (at order Λ^0), and the **Higgs sector with its Yukawa couplings** (also at order Λ^0). It depends on inputs from all three lineages — A from END, G_J from AUT, (H, D, γ, J) from $\langle \cdot, \cdot \rangle$. The lineages do not merge; the spectral action simply *uses* their outputs as inputs. [CLAIMED — proved in fragments, total construction not yet end-to-end]

Concrete numbers that fall out as predictions (a sample, not the full list):

- Higgs mass ≈ 125 GeV [PREDICTED]
- W boson mass ≈ 80.4 GeV [PREDICTED]
- Z boson mass ≈ 91.2 GeV [PREDICTED]
- Hubble constant $H_0 \approx 67.4$ km/s/Mpc (CMB-aligned) [PREDICTED]
- $\sigma_8 \approx 0.81$ [PREDICTED]
- $A_s \approx 2.1 \times 10^{-9}$ [PREDICTED]
- $\delta_{CP} \approx -\pi/2$ (maximal CP violation in the neutrino sector) [PREDICTED]

- $\Sigma m_\nu \approx 0.06$ eV (sum of neutrino masses) [PREDICTED]
- $\alpha_s(M_Z) \approx 0.118$ [PREDICTED]
- Dark energy equation of state $w = -1$ *exactly* [PREDICTED]
- Proton decay lifetime $\tau \sim 10^{35-36}$ years via $p \rightarrow e^+ \pi^0$ [PREDICTED]

These are not parameters that were dialled to fit observation. They emerge from the cascade structure plus standard cited theorems. They are falsifiable — every number above has an experimental verdict either already in hand or accessible to current or near-future instruments.

6. Predictions and Open Problems

The **central prediction** of the theory is structural rather than numerical:

Every mathematical structure that models reality should be reachable, as a lineage, from $M_2(\mathbb{C})$. There may be intermediate generators along the way (Pati-Salam is one). But every part of physics or mathematics that describes anything real should trace, by some path through the cascade and the three lineages, back to the seed.

This is not a tautology. There are competing ToE candidates — string theory, loop quantum gravity, causal sets, ER=EPR — that do not start from a forced minimal seed and do not propose to *derive* the Standard Model gauge group from a uniqueness argument at all. (Whether this programme’s own derivation succeeds in full strict form is still partly open, per §5.) If any field of physics or mathematics that genuinely describes nature turns out to have *no* lineage to $M_2(\mathbb{C})$, this theory is false. That is the falsification criterion.

Beyond that, several **specific falsifiable forks** are open right now:

- **Strong CP problem.** Two paths exist: (A) Pati-Salam left-right symmetry forces $\theta = 0$ with no axion needed, or (B) a QCD axion at mass $\sim 10^{-5}$ to 10^{-3} eV. The two paths are mutually exclusive. **Detect an axion $\rightarrow$ Path B; null at the predicted mass window $\rightarrow$ Path A.**
- **Hubble tension.** The theory predicts the CMB value (≈ 67.4 km/s/Mpc) from the cascade-fixed Newton’s G . Late-time distance-ladder measurements suggesting ≈ 73 must therefore be either (i) a systematic in the calibration, or (ii) a signal of a missing sub-branch (likely early-dark-energy physics not yet placed in the tree).
- **σ_8 tension.** Same shape of fork: theory predicts the Planck value; weak-lensing low- σ_8 must be systematic or a missing late-time branch.
- **Muon g-2.** The theory predicts the SM-aligned value. The Fermilab/BNL excess interpreted as hadronic systematics rather than new physics.

- **B-meson R_K** . Predicted to be SM-aligned ($R_K \rightarrow 1.00$).
- **Dark matter** as sterile neutrino with relic abundance $\Omega_{DM} \approx 0.27$ and direct-detection cross-section set by the cascade couplings.

There are also **open problems** — places where the AI collaboration could not (yet) close the gap. These are named explicitly:

- **Yang-Mills mass gap on $\mathbb{R}^4$** . [CLAIMED, not PROVED.] We have a Bakry-Émery spectral gap of $2/\Lambda^2$ for the restricted quadratic-potential case (PARTIAL — established in Lean for the compact/quadratic setting; the extension to non-compact $\mathbb{R}^4$ is exactly where the project paused), a transfer-matrix $\rightarrow$ mass-gap argument (PARTIAL), and a product-geometry gap transfer (CLAIMED). The full end-to-end argument is not yet airtight.
- **Reflection positivity (OS2)**. [PREDICTED, not yet proved.] Required for the Osterwalder–Schrader reconstruction $\rightarrow$ Wightman QFT path.
- **GNS reconstruction at sufficient detail**. [CLAIMED.] Needed for the full operator-algebraic completion.
- **L^2 spectral theory** for the relevant non-compact operators. [PREDICTED.]
- **Cascade-depth selection** — why physics sits at M_{256} rather than deeper or shallower. [META, OPEN.]

These open problems are not embarrassments. They are the **Mendeleev gaps** — predictions about exactly where the missing piece must live, by the structure of the surrounding theory. The most likely outcome is that they get filled in by the same kind of work that closed the gaps Mendeleev left in 1869: someone with the right instruments goes and looks.

7. The Tree of Reality

All of the above — every node, every lineage, every prediction, every gap — is laid out in one place as the **Tree of Reality**, a phylogenetic cladogram of physics.

The tree is not a file organiser. It is the model of physics itself. Every node is a *claim about reality* (not a category for documents). Every node has exactly one parent. Branches never merge once split — the same rule that holds in biological evolution. Where multiple lineages produce structurally similar results, the similarity is annotated as a **dotted overlay**, never as a merger node.

The status taxonomy is the honesty mechanism. Every claim in the tree wears one of seven tags:

- [PROVED] — unconditional o-sorry Mathlib proof exists
- [PARTIAL] — some evidence, gaps stated explicitly

- **[CLAIMED]** — weak evidence or scaffolding only
- **[PREDICTED]** — no evidence yet, but the structure of the tree forces a specific position; falsifiable
- **[SPECULATIVE]** — an alternative branch the meta-framework permits but does not force
- **[DOWNSTREAM]** — emergent from upstream physics by composition or statistics, not by direct algebraic forcing (chemistry, biology, etc.)
- **[META]** — outside the framework’s deductive reach as currently formulated

Nothing is hidden behind hedged language. If a claim is **[CLAIMED]**, the tree says so out loud.

The tree is open. New lineages may speciate from any node — from the seed, from the cascade, from any stable algebra. Existing branches are predictions about where new fossils should sit; they do not foreclose new lineages. The map is never finished. This is the same posture biology takes towards its own tree of life: it grows.

The full Tree of Reality (v4.3) is published separately as a content-locked specification and will appear as a wing on infinitography.com. It contains roughly several hundred nodes covering everything from the seed through cosmology, condensed matter, and biology — including the open problems sitting at their exact causal positions.

8. How to Verify

This entire programme is designed to be checkable end-to-end by anyone with internet access and a Lean installation. The provenance trail:

- **Papers on Zenodo.** Every paper carries a permanent concept DOI on Zenodo, CERN’s open-access research repository. The 22 papers span the original Infinitography series (1–15), the Gnosis series (16–19), and the synthesis papers (A, B, C). Paper D (machine-verified ToE backbone) is at [10.5281/zenodo.20011540](https://doi.org/10.5281/zenodo.20011540). Paper E (Standard Model emergence) is at [10.5281/zenodo.20012234](https://doi.org/10.5281/zenodo.20012234). Pansophia is at [10.5281/zenodo.19974680](https://doi.org/10.5281/zenodo.19974680).
- **Bitcoin timestamps.** Every paper version, every Lean commit, every site change is hashed and timestamped on the Bitcoin blockchain via OpenTimestamps. The proofs are independently verifiable using the open-source `ots` client — no trust in me or in any service required.
- **Open-source Lean infrastructure.** The full codebase lives at github.com/wonderben-code/convergence-codex. MIT-licensed. `lake build` compiles the lot. The honest picture, as of this writing: ~90 files, thousands of build jobs, 0 outstanding `sorry` in the main libraries. *Critically*, a sizeable fraction of the compiled theorems are scaffolding (type-level identities, arithmetic encodings, dimension-counting wrappers) rather than full mathematical proofs of the physics claims they are named for. The genuinely Mathlib-

backed core — Lawvere fixed-point construction, seed selection, traceless-matrix Lie algebra dimensions via rank-nullity, Clifford-matrix isomorphism, finite Hilbert structure, Kronecker decomposition, the Pati-Salam dimensional embedding — is solid and re-checkable. The remainder is honest work-in-progress, named with status tags rather than hidden. Anyone running `lake build` sees both layers.

- **Code provenance.** The discovery system (Gnosis), formalisation system (Logos), and supporting infrastructure are all in the same repository, MIT-licensed, runnable.

If you want to audit a single claim: find its node in the Tree of Reality, follow the citation to the paper section, follow the section to the Lean file, run `lake build`, watch it compile. Every step is open.

This is the contract. The work has to stand on the verification, not on my word.

9. Honest Limits

What this is:

- A research programme of one human plus frontier AI, producing 22 papers and roughly 200 machine-verified theorems.
- A structural proposal — $\text{Nothing} \rightarrow D = (D \rightarrow D) \rightarrow M_2(\mathbb{C}) \rightarrow \text{three lineages} \rightarrow \text{SM} + \text{GR} + \text{QM}$ — with specific numerical predictions falling out at the end.
- Open, falsifiable, Bitcoin-timestamped, MIT-licensed.

What this is not:

- Peer-reviewed publication. This is preprint-grade independent work. Citations should treat it accordingly.
- Free of AI error. The AI collaborator hallucinates. Some claims in this programme — particularly older ones, particularly informal-prose framings — will turn out to be wrong. The status taxonomy exists precisely to expose where evidence is weak.
- Closed. The tree is open at every level. New branches may speciate. Open problems may have answers we have not yet imagined.

If you find an error, please tell me. The point of doing this work in the open is to be checked. Corrections will be published with attribution and Bitcoin-timestamped like every other revision.

10. Where Next

In rough order of priority:

1. **Upgrade scaffolded Lean theorems to genuine Mathlib proofs.** The §2 honesty about arithmetic-proxy theorems is the most pressing structural debt in the programme. Every downstream physical claim currently sitting on a type-level wrapper deserves to be either elevated to a real proof or downgraded in tag. This is patient line-by-line work, well suited to a human mathematician + AI pair, and it is the first thing the programme needs.
2. **Close the genuinely open problems.** Yang-Mills mass gap on $\mathbb{R}^4$, reflection positivity, full GNS reconstruction, L^2 spectral theory for the non-compact case. These are not random open problems — they are the specific gaps in the chain from $M_2(\mathbb{C})$ to physical observables. Each one closed tightens the cascade. Capable AI mathematicians, or capable human ones, welcome.
3. **Stage B: the 81-field Gnosis run.** Gnosis v2 has been built and tested at small scale. A full run over 81 fields of science and mathematics — the largest convergence survey attempted — will either find new structural patterns supporting the tree, or surface counter-examples we need to account for.
4. **Outreach.** Send this paper and the Tree of Reality to physicists and mathematicians who can audit. The list is large and being curated. If you are one of them and have read this far: thank you. Email is welcomed.
5. **Speciation.** Wherever the tree grows new branches — new physics discovered experimentally, new fields of mathematics with no current lineage — see whether they trace back to $M_2(\mathbb{C})$. If they do, the theory absorbs them. If they don't, the theory has a problem, and we want to know where exactly.

The work doesn't end. It speciates.

Acknowledgements

Built in collaboration with Anthropic's Claude family of models — without which this programme would not exist. The architecture (Gnosis, Logos, Synthesis, Praxis) is custom-built on top of Claude's API and Claude Code, with all design decisions, mathematical direction, and authorship attributable to a single human author.

Every theorem, every paper, every line of code is open source under MIT or CC-BY, and every revision is Bitcoin-timestamped via OpenTimestamps.

References (selected)

- **Pansophia: A Four-Component Architecture for Autonomous Knowledge Work.** DOI: [10.5281/zenodo.19974680](https://doi.org/10.5281/zenodo.19974680)
- **Paper D: Machine-Verified Backbone of the ToE.** DOI: [10.5281/zenodo.20011540](https://doi.org/10.5281/zenodo.20011540)
- **Paper E: Emergence of the Standard Model from $M_2(\mathbb{C})$.** DOI: [10.5281/zenodo.20012234](https://doi.org/10.5281/zenodo.20012234)
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The complete reference set, with every cited theorem tied to its specific node in the Tree of Reality, is maintained at github.com/wonderben-code/convergence-codex.

Independent research, AI-collaborative, not peer-reviewed. Read in that light. Verify what catches your eye. Challenge what you doubt. Tell me if I'm wrong.